AI OPEN
DATA CENTER



Evaluating Heat Rejection Systems Configurations for AI Factories: A Multi-Metric Analysis

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OCP PLATINUM® Member

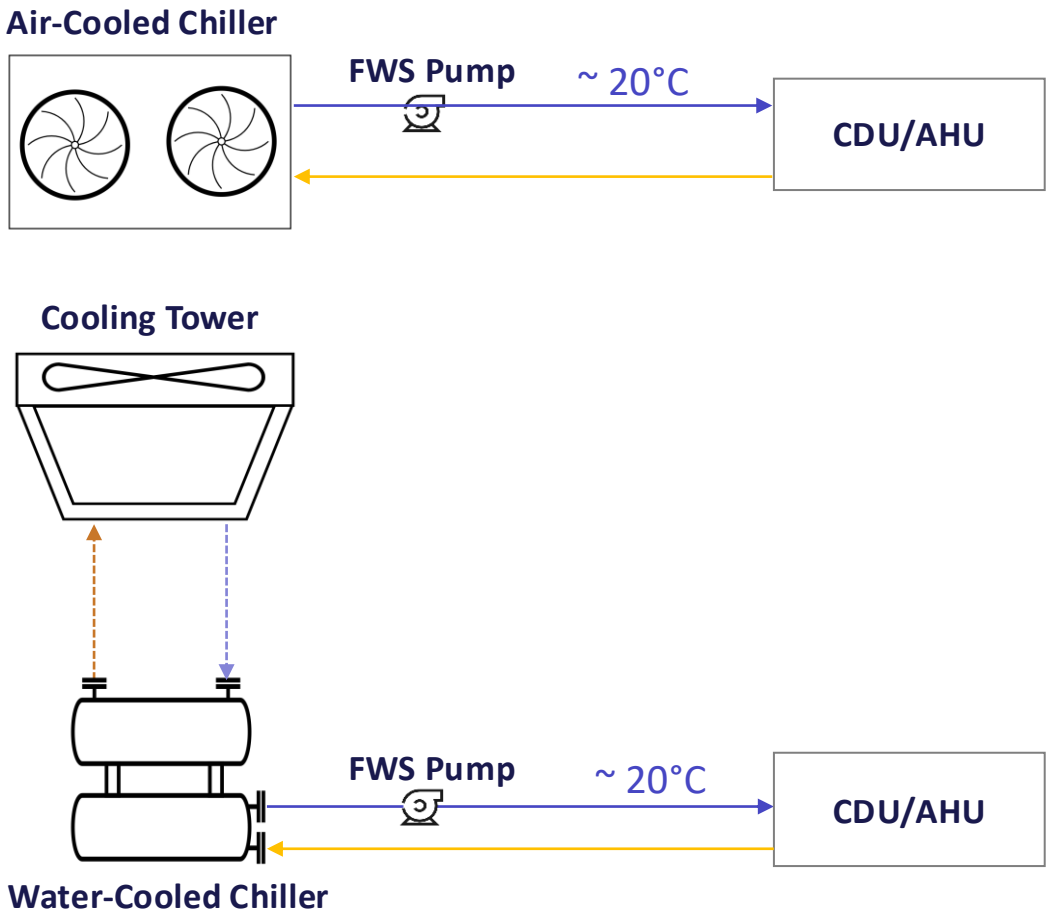
From Data Centers to AI Factories

- Facility cooling is the primary driver of data center efficiency
  - Dominates PUE
  - Direct impact on power available for compute/token generation
- AI factories require rethinking facility design
  - Legacy designs: low-temperature chillers + cooling towers = high power/water use
  - New approach: high-temperature cooling (S45) enables efficiency gains
  - Potential to shift power from cooling to compute
- No single optimal facility design
  - Optimum depends on: Site climate, power/water availability, footprint, TCO, noise, carbon footprint, time to market, …
  - Right design is site-specific and purpose-driven

Facility Water System Design – Legacy DC

Shared FWS Loop:

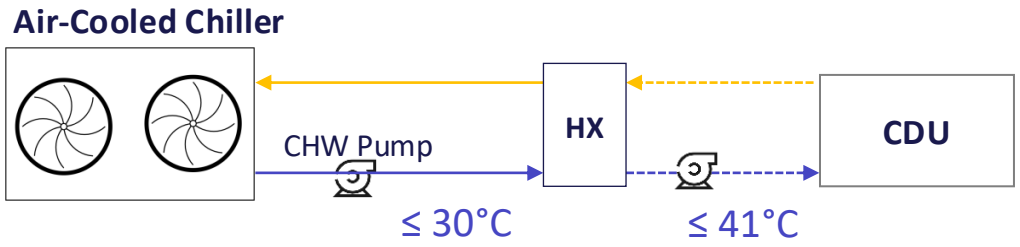
- Similar supply temperature
- Forced to operate at low supply temperature to satisfy all equipment
- Results in excessive cooling power consumption
- High water consumption

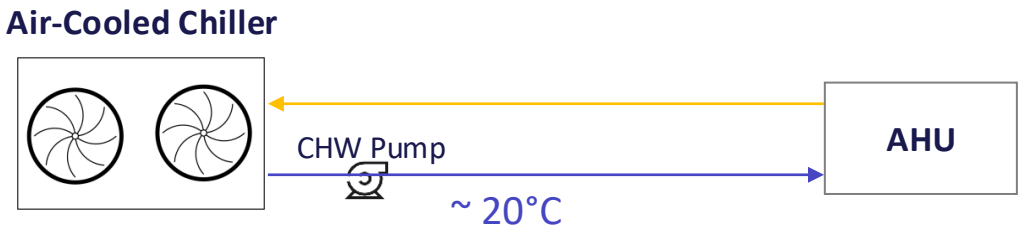


Dedicated Facility Loops

Dedicated FWS Loop:

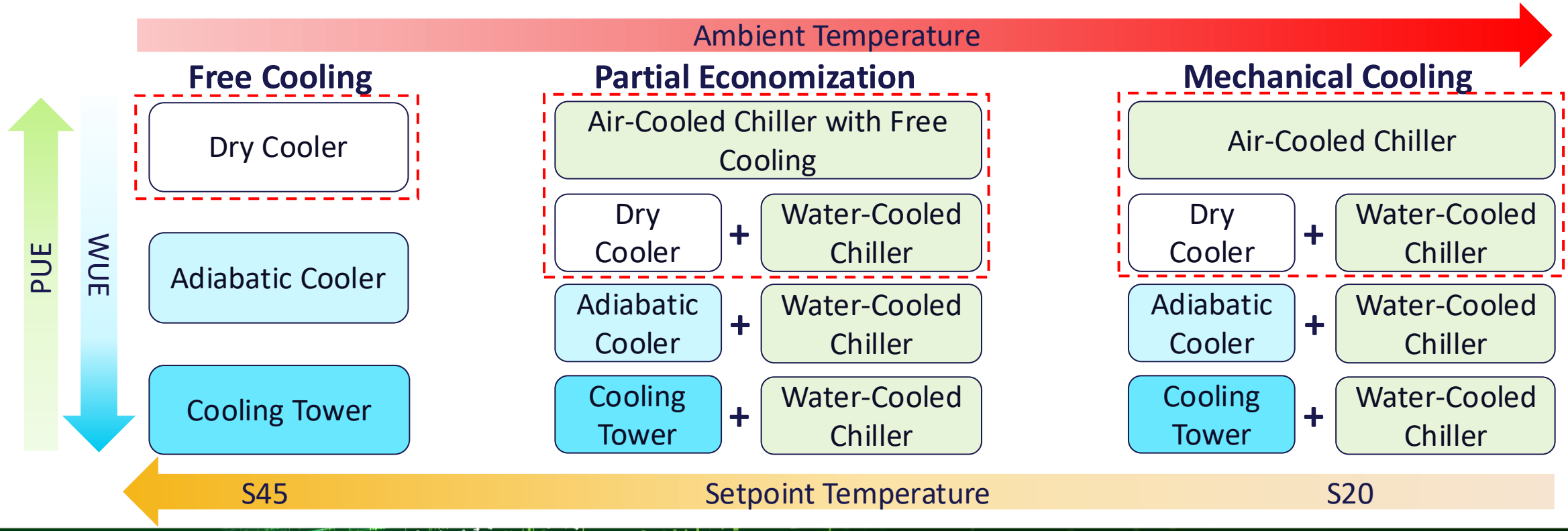
- Transition to liquid cooling enables higher FWS inlet temperatures
- Different equipment, different supply temperatures (separate optimization)
- Higher efficiency
- Better PUE



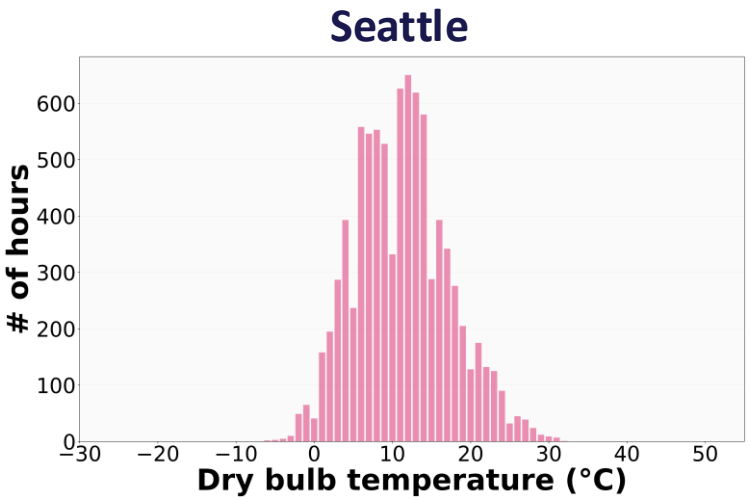


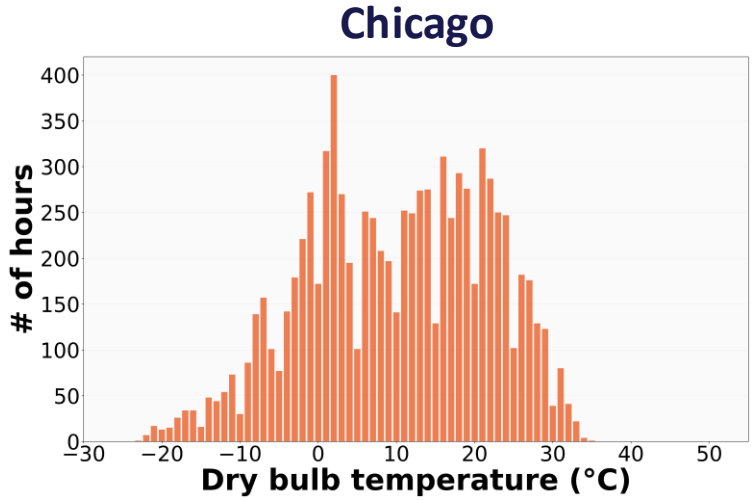
High-Temperature Cooling (S45)

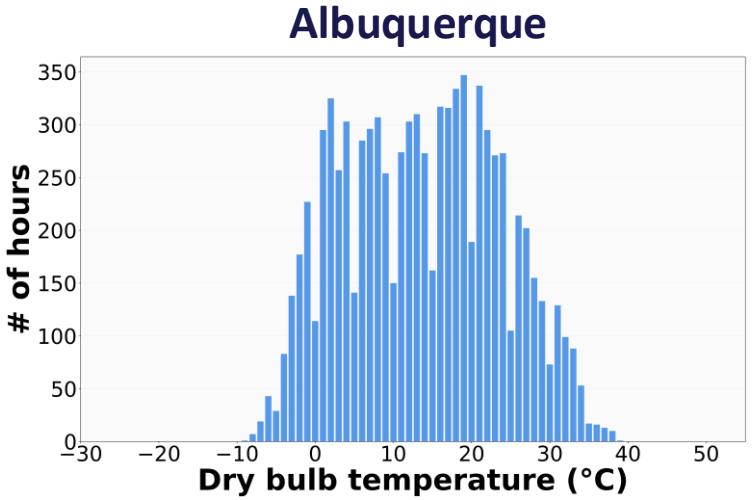
- **Further expands free cooling window** — more hours/seasons where mechanical cooling is unnecessary
- **Mechanical cooling becomes "trim only"** — handles peak conditions only
- **Aggressive free cooling strategy** — leverages outdoor air when available

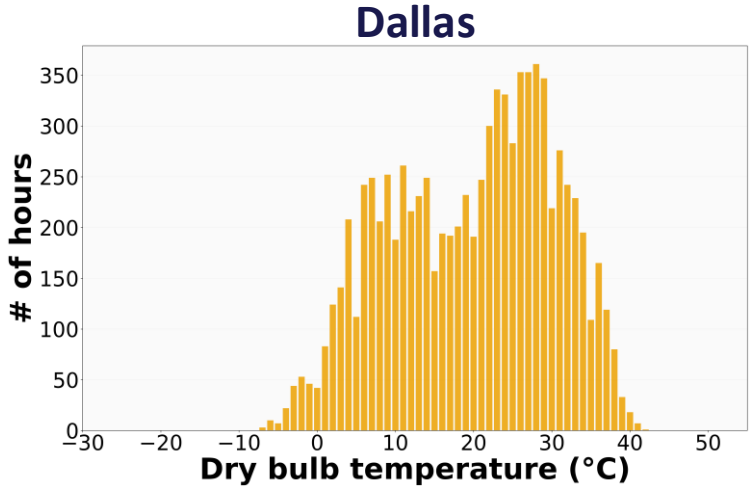


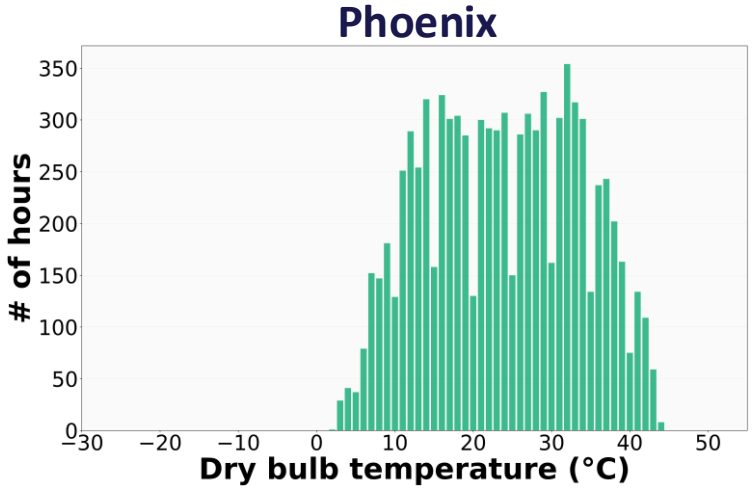
Locations (DB and Elevation)

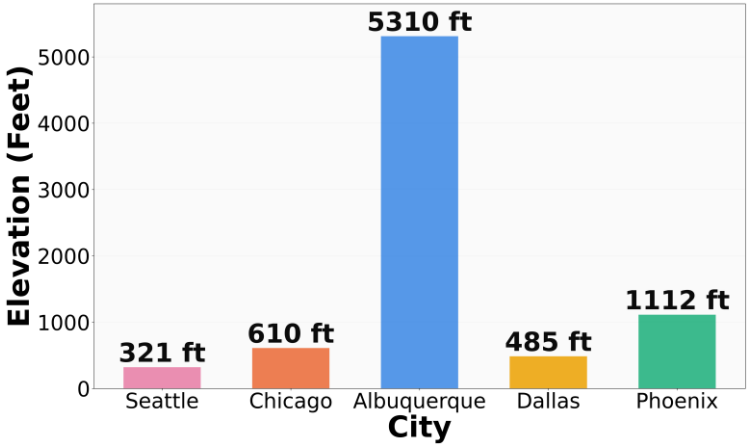










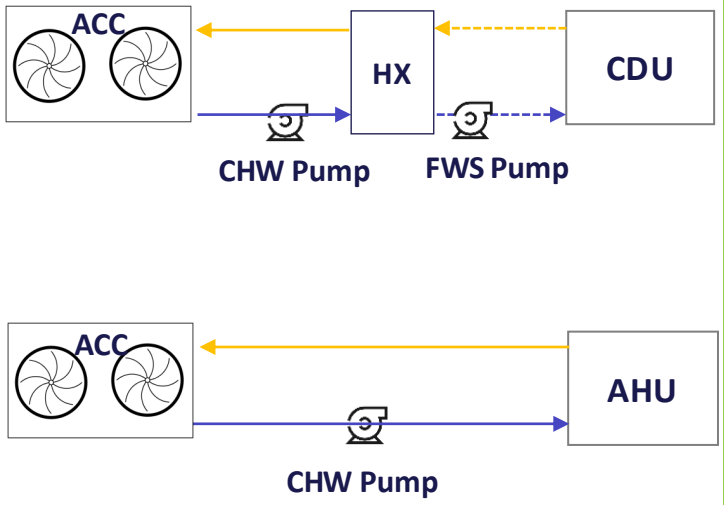


Facility Water System Design – Studied Cases

For a 250 MW AI Factory:

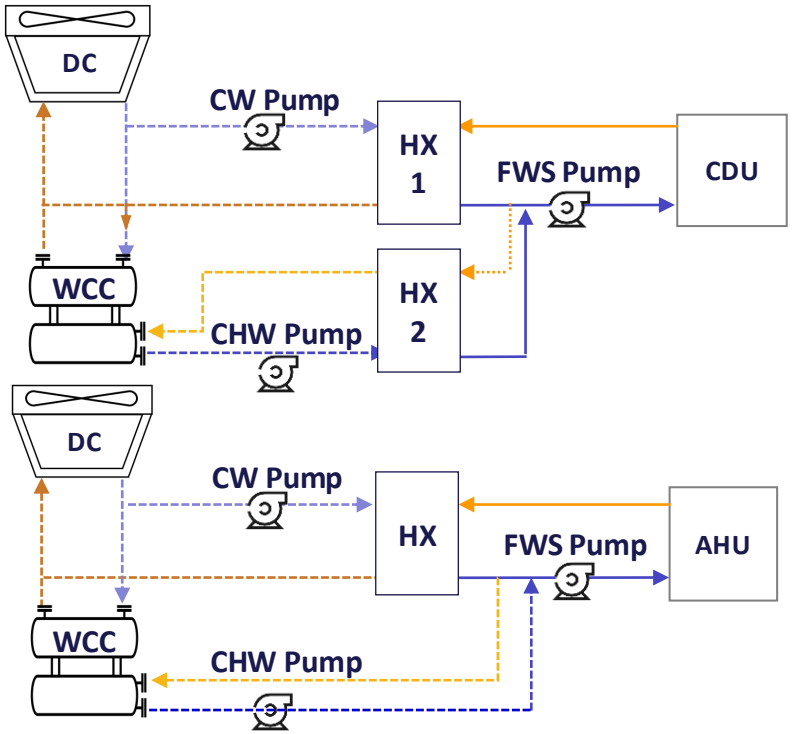
Design 1 - Baseline

- Dedicated loops
- Air-cooled chiller (ACC)



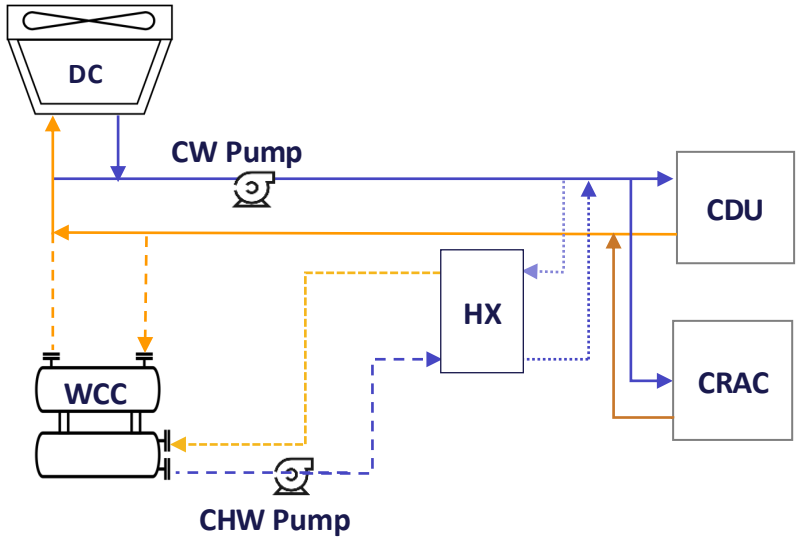
Design 2

- Dedicated loops
- Water-cooled chiller (WCC) + Dry cooler (DC)

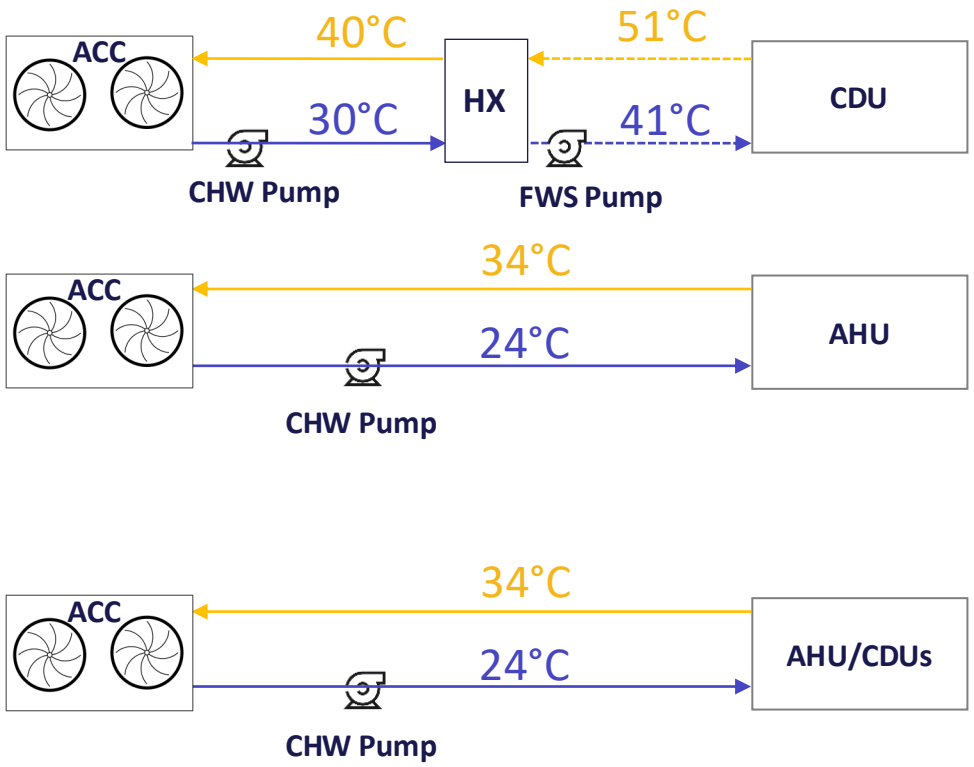


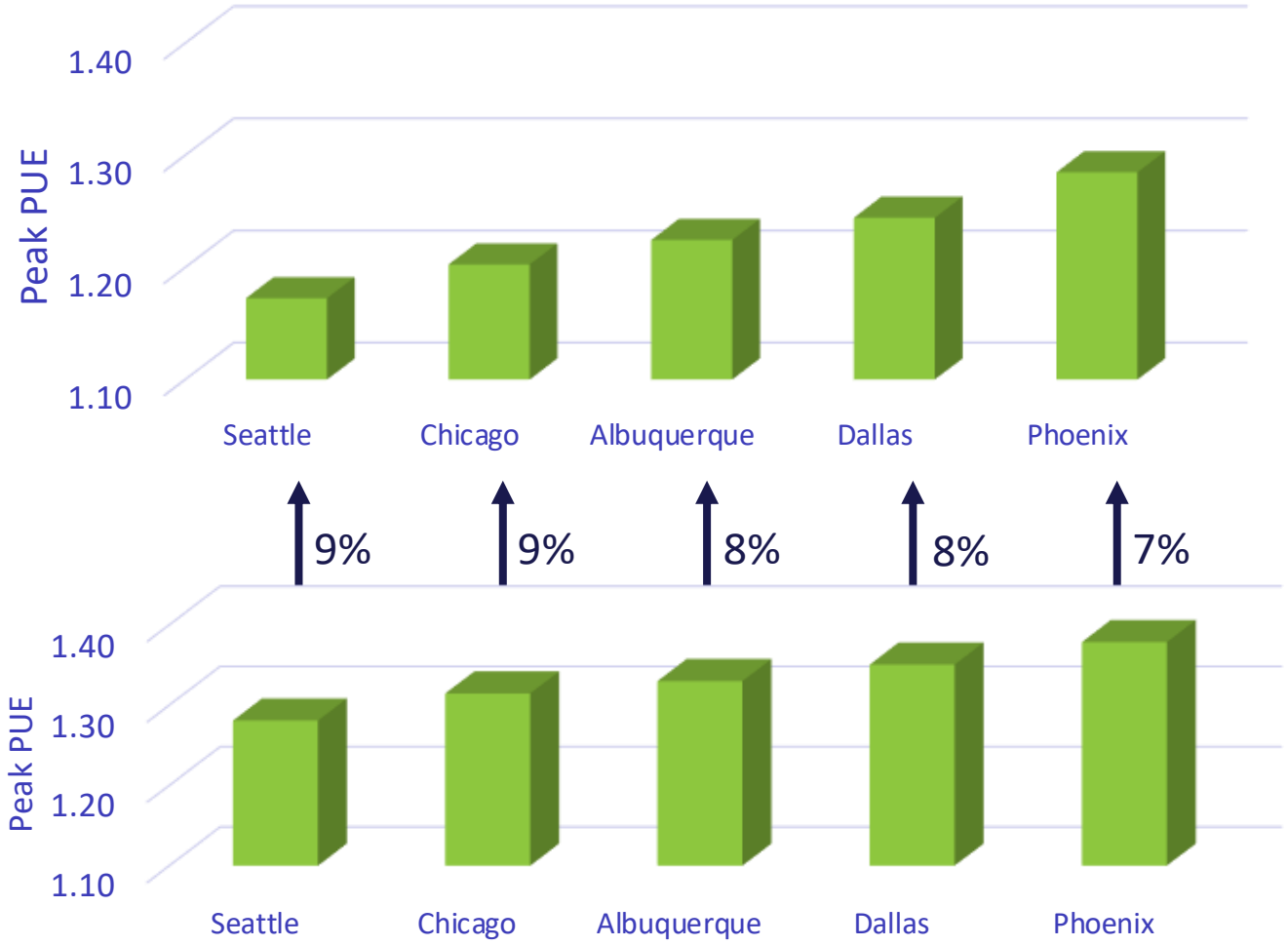
Design 3

- Shared loop
- WCC + DC
- Replacing CRAH/fan walls with CRAC units



PUE Data - Design 1 (Baseline)

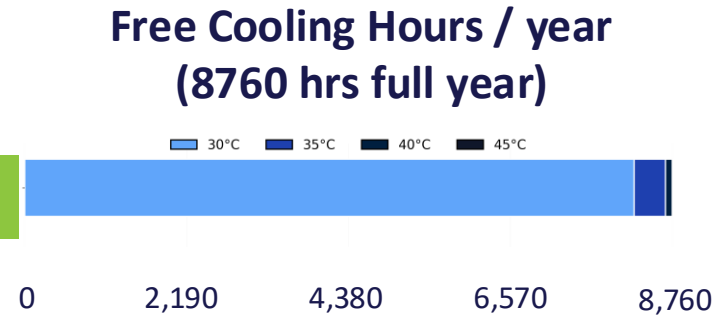


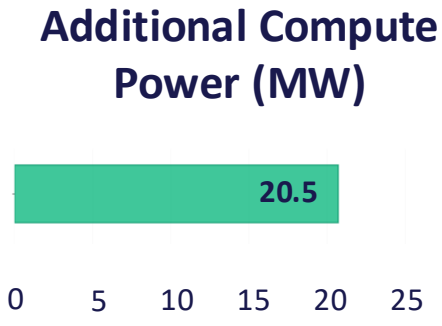


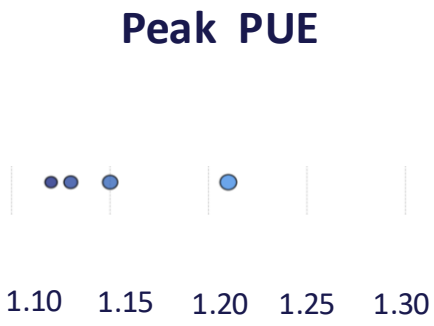
- Transition from shared loop to dedicated loops significantly improves the power efficiency

PUE Data - Design 2

City	30°C	35°C	40°C	45°C
Seattle	Hybrid	Hybrid	Free	Free

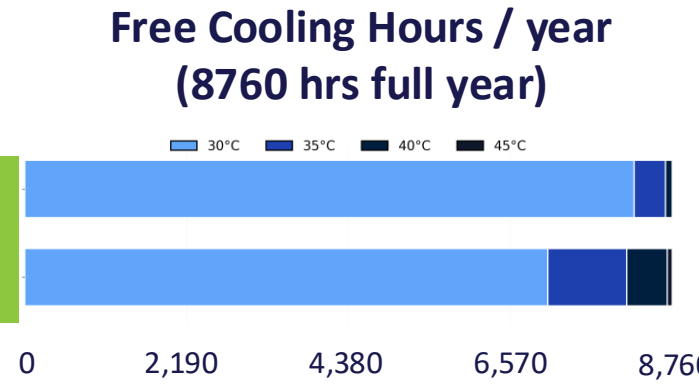


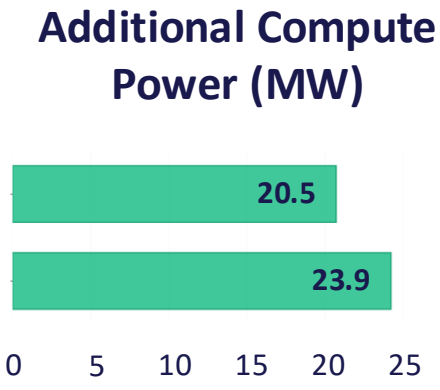


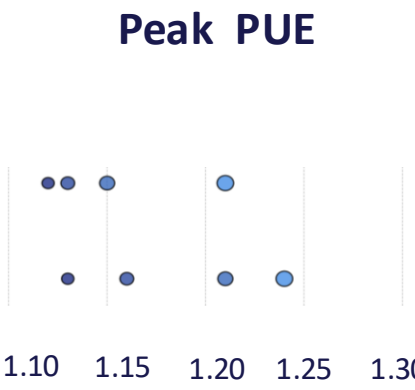


PUE Data - Design 2

City	30°C	35°C	40°C	45°C
Seattle	Hybrid	Hybrid	Free	Free
Chicago	Mech	Hybrid	Hybrid	Free

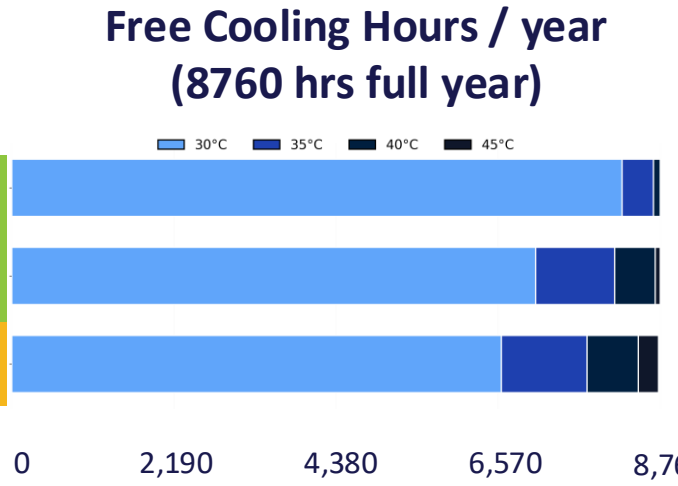


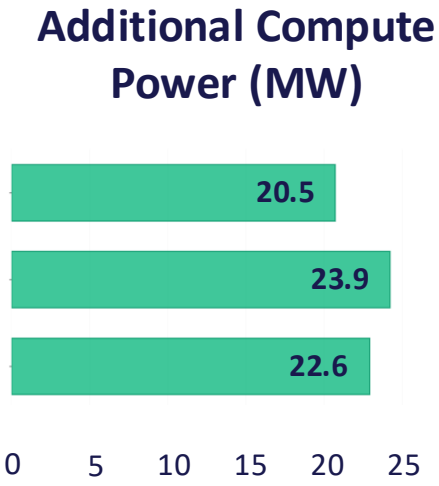


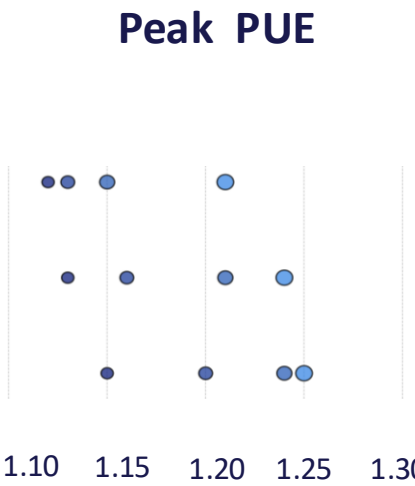


PUE Data - Design 2

City	30°C	35°C	40°C	45°C
Seattle	Hybrid	Hybrid	Free	Free
Chicago	Mech	Hybrid	Hybrid	Free
Albuquerque	Mech	Mech	Hybrid	Hybrid

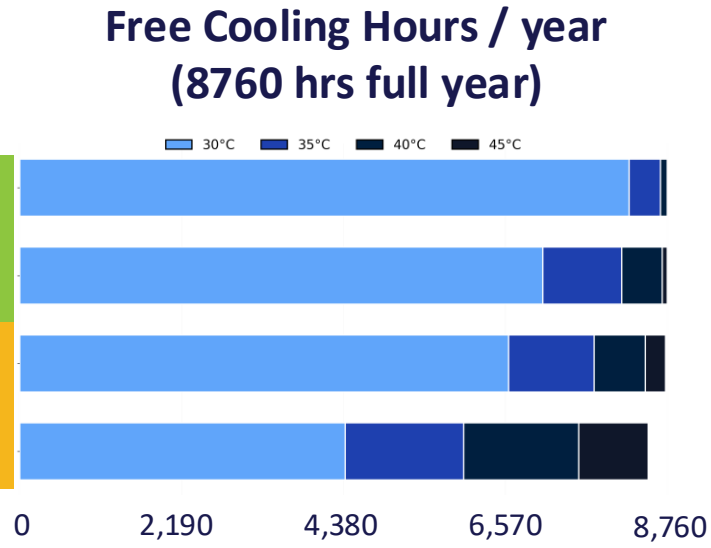


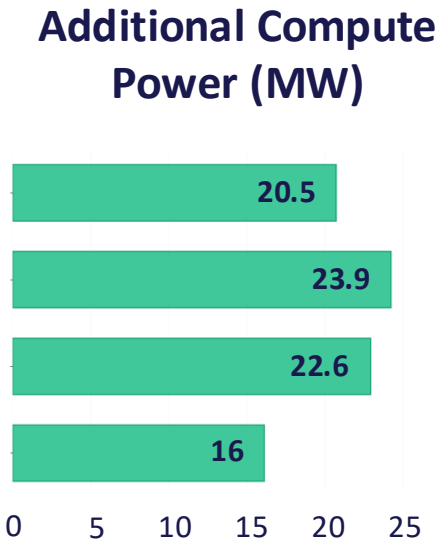


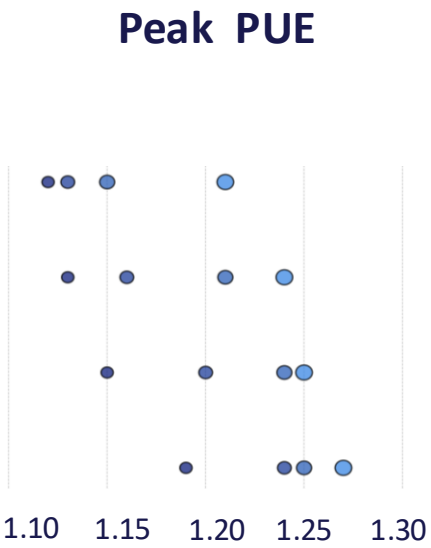


PUE Data - Design 2

City	30°C	35°C	40°C	45°C
Seattle	Hybrid	Hybrid	Free	Free
Chicago	Mech	Hybrid	Hybrid	Free
Albuquerque	Mech	Mech	Hybrid	Hybrid
Dallas	Mech	Mech	Hybrid	Hybrid

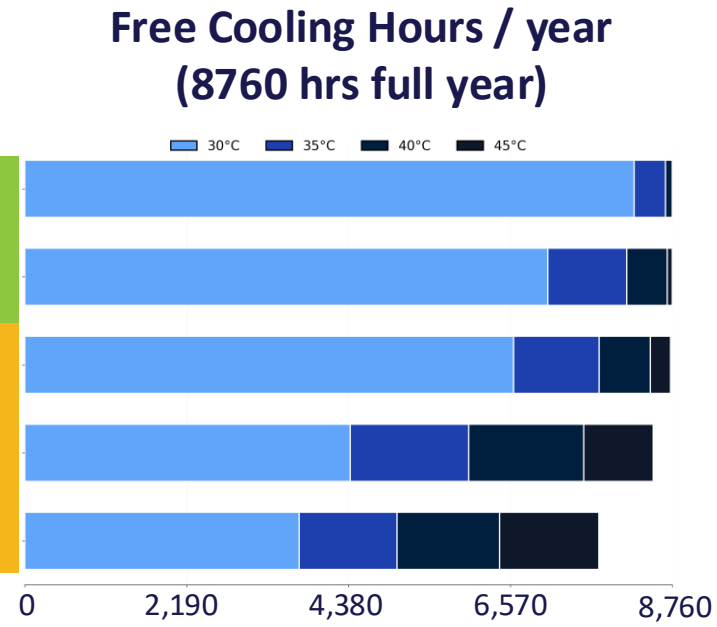


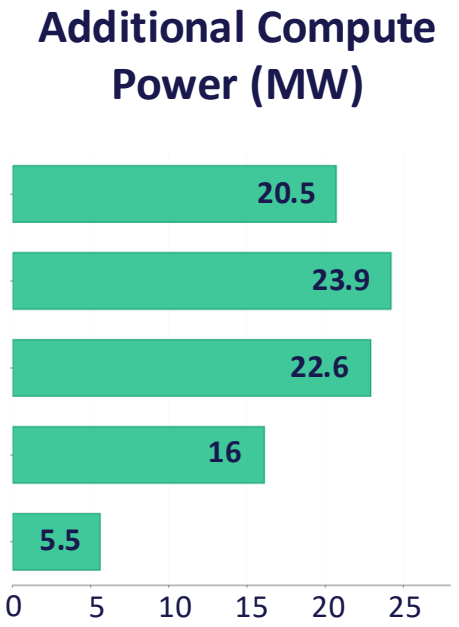


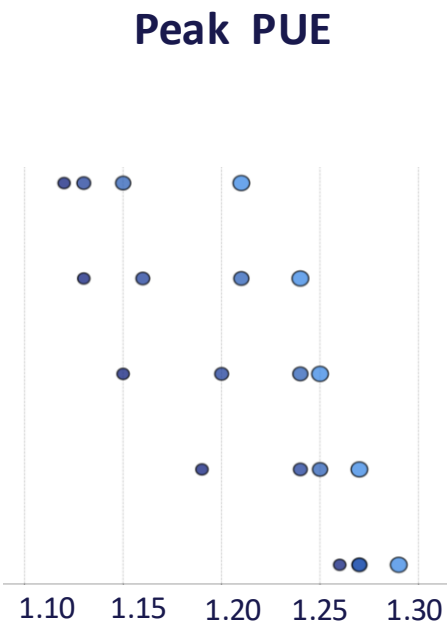


PUE Data - Design 2

City	30°C	35°C	40°C	45°C
Seattle	Hybrid	Hybrid	Free	Free
Chicago	Mech	Hybrid	Hybrid	Free
Albuquerque	Mech	Mech	Hybrid	Hybrid
Dallas	Mech	Mech	Hybrid	Hybrid
Phoenix	Mech	Mech	Mech	Hybrid

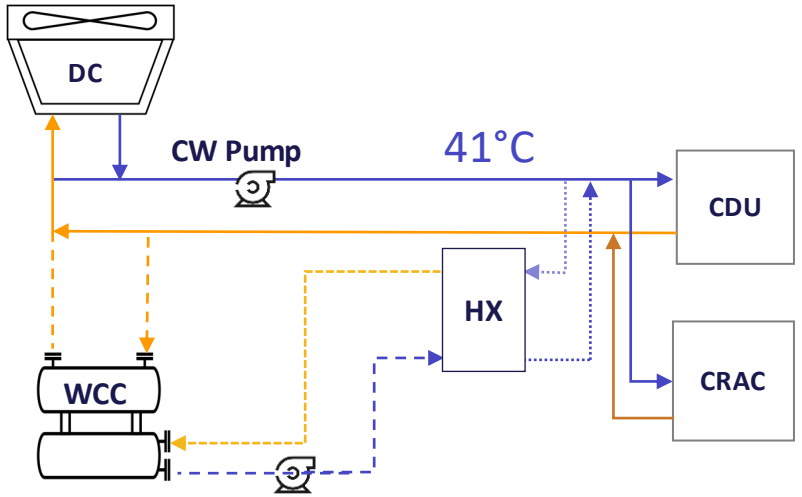




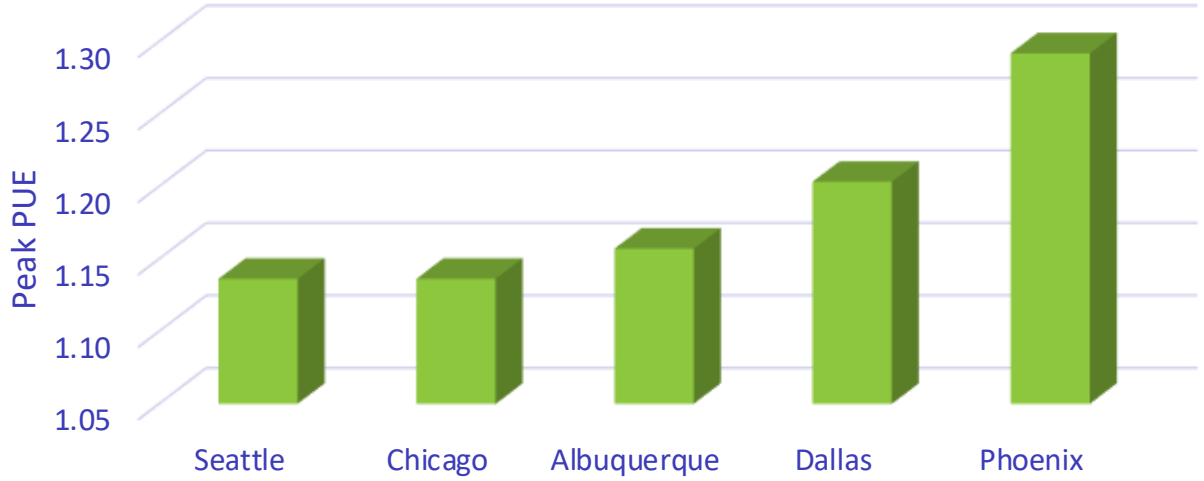


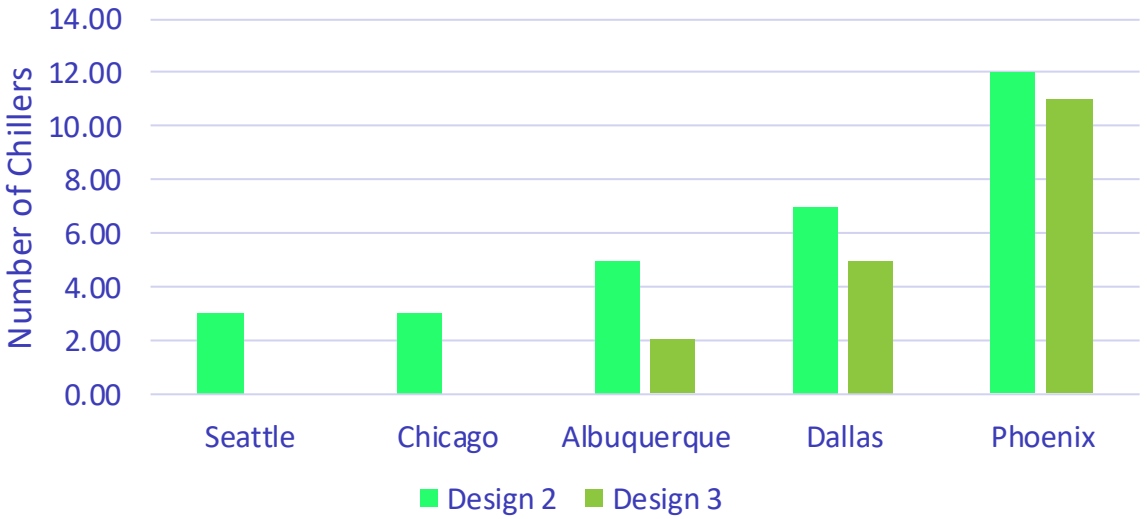
- Zero water consumption
- No full mechanical cooling at S45
- Significant increase in compute power across various locations by moving from S30 to S45
- Up to 46% and 10% drop in facility power consumption in Chicago and Phoenix, respectively

PUE Data – Design 3



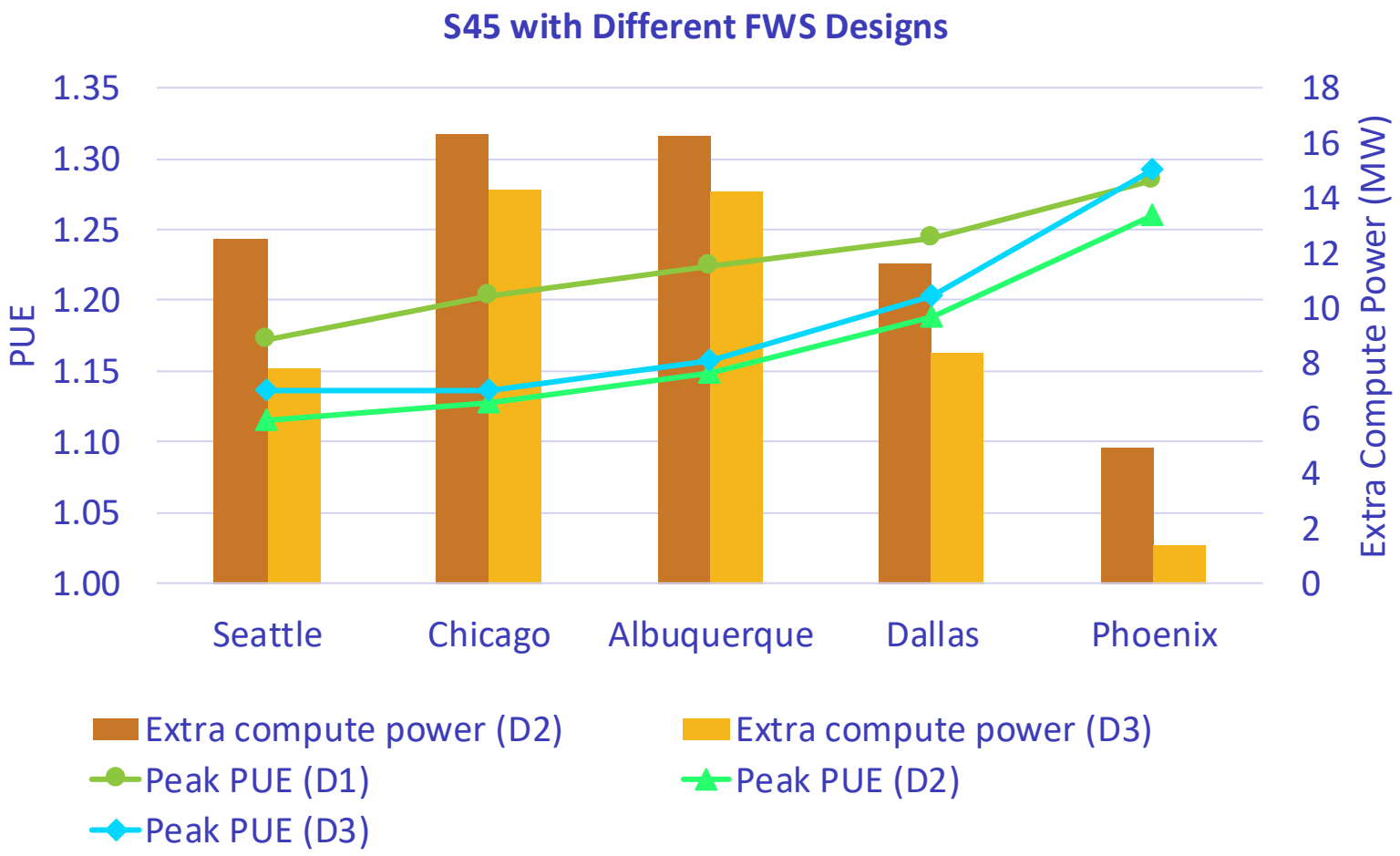
- Replacing CRAH/fan wall units with CRAC systems to enable high temperature shared loop
- Reduced piping
- Less number of chillers compared to design 2





S45 Across Different Designs

- Highest energy efficiency with design 2
- Up to 16 MW (37%) and 14 MW (33%) higher compute power in design 2 and 3 vs design 1
- Higher efficiency → Higher compute power → Higher token generation → Higher revenue
- Zero water consumption in all three designs



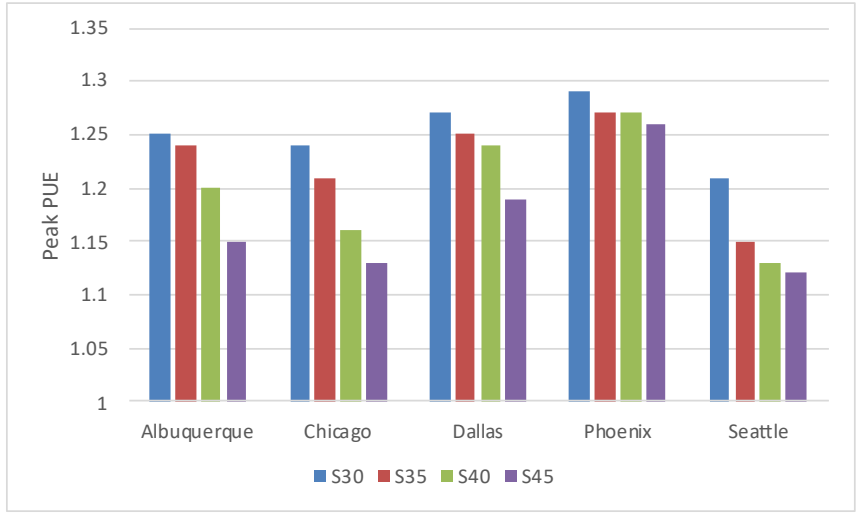
TCO Analysis (S45)

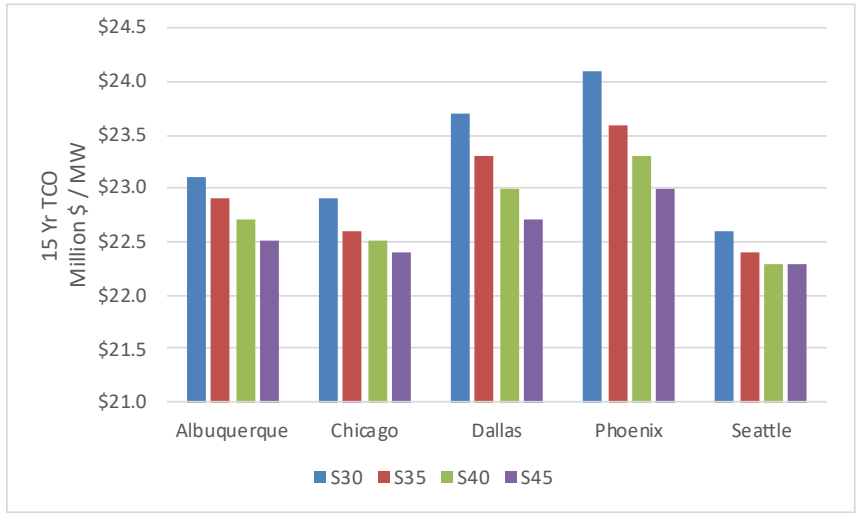
City	HR Plant	Peak PUE	Annual PUE	CapEx (1) (per MW)	Annual OpEx (2) (per MW)	15 Yr TCO (1)(2) (per MW)	HR Plant Area
Seattle	Design 1	1.17	1.13	$1.42M	$1.19M	$23.6M	230K SF
	Design 2	1.12	1.09	$0.93M	$1.14M	$22.3M	100K SF
	Design 3	1.14	1.12	$0.82M	$1.17M	$22.8M	100K SF
Chicago	Design 1	1.20	1.14	$1.42M	$1.19M	$23.7M	230K SF
	Design 2	1.13	1.09	$0.94M	$1.15M	$22.4M	100K SF
	Design 3	1.14	1.12	$0.82M	$1.18M	$22.8M	100K SF
Albuquerque	Design 1	1.22	1.14	$1.52M	$1.20M	$23.9M	250K SF
	Design 2	1.15	1.10	$1.01M	$1.15M	$22.5M	110K SF
	Design 3	1.16	1.12	$0.94M	$1.18M	$22.9M	110K SF
Dallas	Design 1	1.24	1.16	$1.49M	$1.22M	$24.2M	245K SF
	Design 2	1.19	1.10	$1.02M	$1.16M	$22.7M	105K SF
	Design 3	1.20	1.12	$0.97M	$1.18M	$23.0M	105K SF
Phoenix	Design 1	1.29	1.17	$1.58M	$1.23M	$24.6M	265K SF
	Design 2	1.26	1.11	$1.14M	$1.17M	$23.0M	115K SF
	Design 3	1.29	1.13	$1.12M	$1.19M	$23.3M	115K SF

(1) CapEx and TCO values only include differences in Mechanical plant variations between Designs 1, 2 & 3
(2) OpEx Assumptions: $0.12/kWh, 3% Inflation, only energy costs are included

TCO Across Different TCS Temperatures

- TCS temperature variations shown for Design 2
- Highest energy efficiency achieved with S45
- Lowest TCO achieved with S45
- Higher efficiency → Higher compute power → Higher token generation → Higher revenue





TCO Analysis (Design 2)

City	TCS Temp	Peak PUE	Annual PUE	CapEx [1] (per MW)	Annual OpEx [2] (per MW)	15 Yr TCO [1][2] (per MW)
Seattle	S30	1.21	1.10	$1.10M	$1.15M	$22.6M
	S35	1.15	1.09	$0.99M	$1.15M	$22.4M
	S40	1.13	1.09	$0.93M	$1.15M	$22.3M
	S45	1.12	1.09	$0.93M	$1.15M	$22.3M
Chicago	S30	1.24	1.11	$1.18M	$1.16M	$22.9M
	S35	1.21	1.10	$1.10M	$1.15M	$22.6M
	S40	1.16	1.09	$1.01M	$1.15M	$22.5M
	S45	1.13	1.09	$0.94M	$1.15M	$22.4M
Albuquerque	S30	1.25	1.12	$1.23M	$1.17M	$23.1M
	S35	1.24	1.10	$1.22M	$1.16M	$22.9M
	S40	1.20	1.10	$1.12M	$1.15M	$22.7M
	S45	1.15	1.10	$1.01M	$1.15M	$22.5M
Dallas	S30	1.27	1.15	$1.20M	$1.20M	$23.7M
	S35	1.25	1.13	$1.20M	$1.18M	$23.3M
	S40	1.24	1.11	$1.17M	$1.17M	$23.0M
	S45	1.19	1.10	$1.02M	$1.16M	$22.7M
Phoenix	S30	1.29	1.17	$1.22M	$1.23M	$24.1M
	S35	1.27	1.14	$1.21M	$1.20M	$23.6M
	S40	1.27	1.13	$1.17M	$1.19M	$23.3M
	S45	1.26	1.11	$1.14M	$1.17M	$23.0M

(1) CapEx and TCO values only include differences in Mechanical plant variations between S30, S35, S40 and S45

(2) OpEx Assumptions: $0.12/kWh, 3% Inflation, only energy costs are included

TCO Key Takeaways

- WCC + DC + CRAH (Design 2) is the most efficient and provides the lowest TCO over a 15-year period for all regions and climates
- WCC + DC + CRAC (Design 3) has the lowest CapEx but slightly higher OpEx than Design 2 so results in an overall higher TCO
- Dry coolers require approximately two thirds of the site area as compared to air cooled chillers
- As TCS temperature increases from S30 to S45, peak PUE and TCO will decrease accordingly, which enables more IT power to be utilized for a fixed utility capacity

Call to Action

- Problem to solve
  - Facility design optimization to improve efficiency of AI factories
  - Reducing cooling power consumption to lower environmental impacts and enable higher compute output
  - Reducing on-site water consumption to zero
  - Consider sound/noise impact on adjacent properties for different heat rejection components
  - Design modularity to support different cooling strategies and future expansion
  - Develop waste heat recovery strategies to repurpose heat for secondary applications

Thank You!

